II Year B.Tech. CE – I Semester

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(23A01301P) STRENGTH OF MATERIALS LAB

Course objectives:

- To determine the tensile strength and yield parameters of mild steel
- To find out flexural strengths of Steel/Wood specimens and measure deflections
- To determine the torsion parameters of mild steel bar
- To determine the hardness numbers, impact and shear strengths of metals
- To determine the load-deflection parameters for springs

Course Outcomes:

- Conduct tensile strength test and draw stress-strain diagrams for ductile metals
- Perform bending test and determine load-deflection curve of steel/wood
- Able to conduct torsion test and determine torsion parameters
- Perform hardness, impact and shear strength tests and calculate hardness numbers, impact and shear strengths
- Able to conduct tests on closely coiled and open coiled springs and calculate deflections

LIST OF EXPERIMENTS:

1. Tension test.
2. Bending test on (Steel/Wood) Cantilever beam.
3. Bending test on simply supported beam.
4. Torsion test.
2. Hardness test.
3. Compression test on Open coiled springs
4. Tension test on Closely coiled springs
5. Compression test on wood/ concrete
6. Izod / Charpy Impact test on metals
7. Shear test on metals
8. Use of electrical resistance strain gauges.
9. Continuous beam – deflection test.